

**CUSTOMER RETENTION AND USAGE INTENSITY
MODELING USING INSTACART ONLINE GROCERY
TRANSACTION DATA**

**DATA 606 FINAL PROJECT REPORT
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1 Introduction

Customer retention is a critical component of business success in online retail platforms, particularly in online grocery services where customer lifetime value depends heavily on repeat purchasing behavior. Unlike one-time transactions, online grocery platforms rely on consistent customer engagement to maintain revenue stability and growth. Customer behavior can be analyzed using transactional data, which captures detailed information about ordering patterns, purchase timing, and engagement duration. By analyzing this data, statistical and machine learning models can be developed to predict customer retention and estimate usage intensity.

However, modeling customer behavior presents statistical challenges. Customer order frequency represents count data and often exhibits over-dispersion, where the variance exceeds the mean. Additionally, customer churn is not directly observed but must be inferred from behavioral patterns such as recency and frequency of purchases. This project uses transactional data from Instacart to model customer retention and usage intensity using statistical methods learned in DATA 606.

Customer churn has significant financial implications for businesses, as acquiring new customers is considerably more expensive than retaining existing ones. Identifying customers at risk of churn enables companies to implement targeted retention strategies, such as personalized promotions or engagement campaigns, which can improve customer loyalty and increase lifetime value.

Similarly, understanding customer usage intensity helps businesses forecast demand, optimize resource allocation, and improve operational efficiency. Therefore, accurate modeling of customer churn and usage intensity provides valuable insights that support data-driven business decision-making

1.1 Purpose

The purpose of this study is to analyze customer ordering behavior and develop predictive models that estimate customer churn and usage intensity.

This project aims to:

- Predict customer churn using behavioral features derived from transactional data
- Model customer usage intensity using count regression models
- Compare statistical and machine learning approaches

- Identify models that provide the best balance between predictive accuracy and interpretability

The results of this study provide insights into customer engagement patterns and demonstrate the application of statistical modeling techniques to real-world transactional data.

1.2 Dataset

The dataset used in this study is the Instacart Online Grocery Basket Analysis dataset obtained from Kaggle. Instacart is an online grocery delivery platform that allows customers to purchase groceries through a digital interface. The dataset contains anonymized transactional records representing customer ordering behavior over time. This dataset provides a real-world example of customer purchasing behavior and is widely used in customer analytics and predictive modeling studies.

The dataset contains approximately 3.4 million orders from 206,209 customers and captures customer engagement over a fixed observation period. Each record in the dataset represents a customer order and includes information about order sequence, order timing, and time between consecutive orders. This information enables analysis of customer engagement patterns, retention behavior, and usage intensity.

For the purposes of this study, the primary dataset used was the orders.csv file, which contains order-level behavioral information. This dataset is particularly suitable for modeling customer retention and engagement because it captures the timing and frequency of customer interactions with the platform.

The orders dataset contains the following variables:

- **user_id**
This variable represents a unique identifier assigned to each customer. It allows tracking of individual customer ordering behavior across multiple transactions.
- **order_number**
This variable represents the sequential number of orders placed by each customer. It indicates the total number of orders placed by a customer and serves as a direct measure of customer engagement and usage intensity.
- **order_dow**
This variable represents the day of the week on which the order was placed, with values ranging from 0 to 6. This variable provides insights into temporal ordering patterns and customer behavior trends.
- **order_hour_of_day**

This variable represents the hour of the day at which the order was placed, with values ranging from 0 to 23. This variable allows analysis of customer ordering habits and engagement timing.

- **days_since_prior_order**
This variable represents the number of days between consecutive customer orders. This is one of the most important behavioral variables in this study, as it allows measurement of customer engagement regularity and recency. Missing values in this variable correspond to customers' first orders, where no prior order exists.

```
Index(['order_id', 'user_id', 'eval_set', 'order_number', 'order_dow',
      'order_hour_of_day', 'days_since_prior_order'],
      dtype='object')
```

	order_id	user_id	eval_set	order_number	order_dow	order_hour_of_day	\
0	2539329	1	prior	1	2	8	
1	2398795	1	prior	2	3	7	
2	473747	1	prior	3	3	12	
3	2254736	1	prior	4	4	7	
4	431534	1	prior	5	4	15	

	days_since_prior_order
0	NaN
1	15.0
2	21.0
3	29.0
4	28.0


```
order_id      0
user_id      0
eval_set      0
order_number  0
order_dow     0
order_hour_of_day  0
days_since_prior_order    206209
dtype: int64
```

Figure 1.2.1: Structure and preview of the Instacart orders dataset

Figure 1.2.1 shows the structure of the orders dataset, including customer identifiers and behavioral timing variables. The presence of sequential ordering information and time intervals between orders makes this dataset particularly suitable for modeling customer retention and usage intensity.

The dataset is appropriate for this study because it contains detailed behavioral information required to construct customer-level features such as order frequency, recency, and engagement duration. These features are essential for predicting customer churn and modeling customer usage intensity using statistical and machine learning methods.

1.3 Methodology

The analysis follows a structured statistical modeling workflow consisting of:

- Data Preprocessing:
Handling missing values and removing unnecessary variables
- Feature Engineering:
Aggregating order-level data to customer-level behavioral features
- Exploratory Data Analysis:
Analyzing behavioral distributions and identifying statistical properties
- Churn Modeling:
Using classification models including Logistic Regression, Decision Tree, and Random Forest
- Usage Intensity Modeling:
Using count regression models including Poisson and Negative Binomial regression
- Model Evaluation
Comparing models using performance metrics including Accuracy, AUC, RMSE, and AIC

The final models were selected based on predictive performance and statistical validity.

1.4 Pre-Processing of the Data

Data preprocessing was performed to prepare the Instacart transactional data for statistical analysis and predictive modeling. This involved examining the structure of the dataset, handling missing values, removing irrelevant variables, and transforming order-level data into customer-level behavioral features.

orders.dtypes	
	0
order_id	int64
user_id	int64
eval_set	object
order_number	int64
order_dow	int64
order_hour_of_day	int64
days_since_prior_order	float64
dtype: object	

Figure 1.4.1: Data Types of Dataset Variables

Figure 1.4.1 shows the data types of each variable in the Instacart orders dataset. The variables `user_id`, `order_number`, `order_dow`, and `order_hour_of_day` are stored as integer data types, indicating discrete numerical values representing customer identifiers and order timing information. These variables are appropriate for statistical analysis because they represent structured behavioral information. The variable `days_since_prior_order` is stored as a floating-point data type, which reflects the presence of missing values in the dataset. Since this variable represents the number of days between consecutive orders, it is considered a continuous numerical variable and plays a key role in measuring customer engagement patterns.

	user_id	order_number	order_dow	order_hour_of_day	days_since_prior_order
count	3.421083e+06	3.421083e+06	3.421083e+06	3.421083e+06	3.421083e+06
mean	1.029782e+05	1.715486e+01	2.776219e+00	1.345202e+01	1.044488e+01
std	5.953372e+04	1.773316e+01	2.046829e+00	4.226088e+00	9.308727e+00
min	1.000000e+00	1.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
25%	5.139400e+04	5.000000e+00	1.000000e+00	1.000000e+01	4.000000e+00
50%	1.026890e+05	1.100000e+01	3.000000e+00	1.300000e+01	7.000000e+00
75%	1.543850e+05	2.300000e+01	5.000000e+00	1.600000e+01	1.500000e+01
max	2.062090e+05	1.000000e+02	6.000000e+00	2.300000e+01	3.000000e+01

Figure 1.4.2: Summary Statistics of Order-Level Data

Figure 1.4.2 shows summary statistics for the order-level dataset, including measures such as mean, minimum, maximum, and standard deviation. The average order_number indicates the typical number of orders placed by customers, while the maximum value shows that some customers place significantly more orders than others.

The variable days_since_prior_order shows the average time between customer orders, indicating general purchasing patterns. The variation in this variable suggests that customers have different engagement levels. These summary statistics provide an overview of customer ordering behavior and confirm that the dataset contains useful behavioral information for modeling.

	user_id	frequency	total_active_days	avg_days_between_orders
0	1	11	190.0	17.272727
1	2	15	228.0	15.200000
2	3	13	144.0	11.076923
3	4	6	85.0	14.166667
4	5	5	46.0	9.200000

Figure 1.4.3: Customer-Level Feature Engineering Through Aggregation

Figure 1.4.3 shows the aggregation of order-level data into customer-level behavioral features using the user_id variable. The frequency variable represents the total number of orders placed by each customer and serves as a key indicator of customer engagement.

	user_id	frequency	total_active_days	avg_days_between_orders	recency	churn
count	206209.000000	206209.000000	206209.000000	206209.000000	206209.000000	206209.000000
mean	103105.000000	16.590367	173.284377	13.428600	17.061782	0.306417
std	59527.555167	16.654774	100.998967	5.496185	10.672178	0.461006
min	1.000000	4.000000	0.000000	0.000000	0.000000	0.000000
25%	51553.000000	6.000000	89.000000	9.000000	7.000000	0.000000
50%	103105.000000	10.000000	151.000000	13.320000	15.000000	0.000000
75%	154657.000000	20.000000	254.000000	17.687500	30.000000	1.000000
max	206209.000000	100.000000	365.000000	27.000000	30.000000	1.000000

Figure 1.4.4: Summary Statistics of Customer-Level Features

Figure 1.4.4 presents summary statistics of the engineered customer-level features. The frequency variable shows variation in customer engagement, with some customers placing many more orders than others. This variation is important for identifying highly engaged and low-engagement customers.

The `total_active_days`, `avg_days_between_orders`, and `recency` variables provide additional information about customer activity patterns. Customers with larger recency values are less active and may be at higher risk of churn. These features are essential inputs for the statistical models used in this study.

1.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution and characteristics of customer behavioral features before building predictive models. This step is important because it helps identify patterns, variability, and potential issues such as skewness, imbalance, and over-dispersion. Understanding these characteristics ensures that appropriate statistical models are selected and that modeling assumptions are valid.

EDA focused on analyzing customer order frequency and churn distribution. These variables are central to the project objectives, as frequency represents usage intensity and churn represents customer retention. Examining the distributions of these variables provides insights into customer engagement patterns and helps guide model selection.

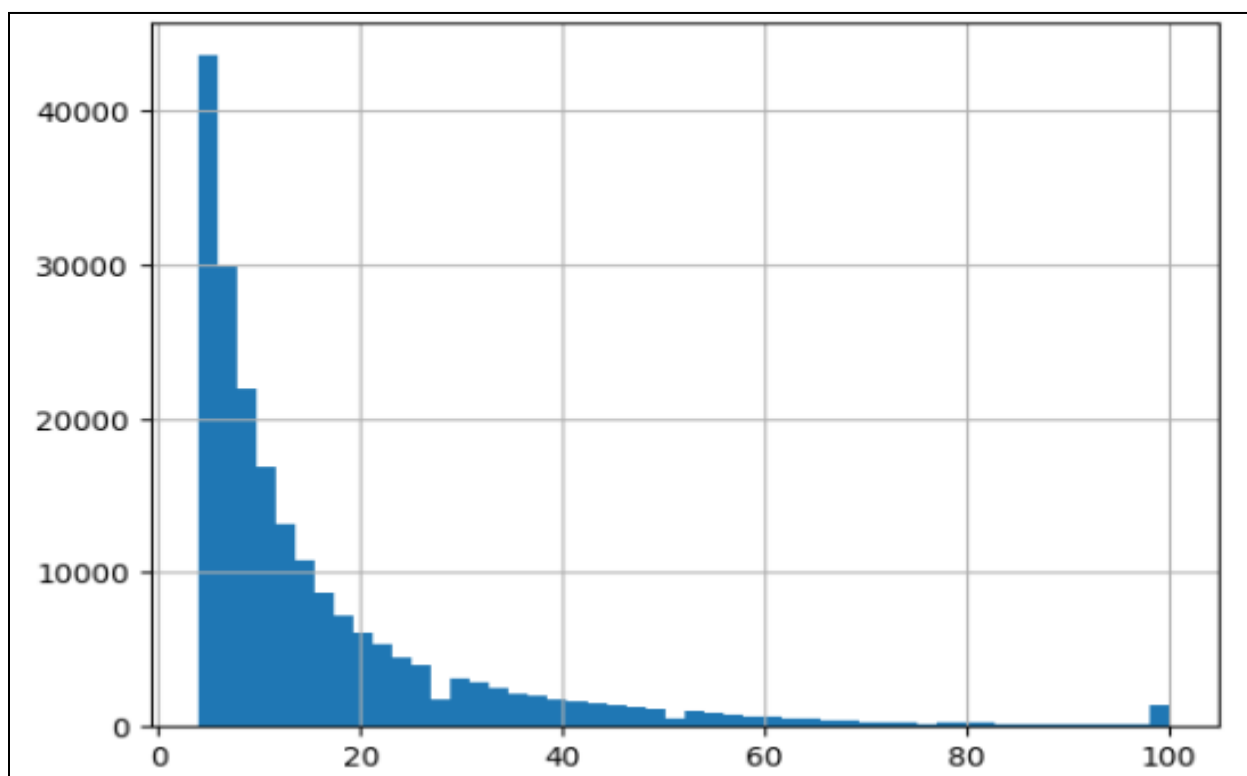


Figure 1.5.1: Distribution of Customer Order Frequency

Figure 1.5.1 shows the distribution of customer order frequency. The distribution is highly right-skewed, indicating that most customers place relatively few orders, while a smaller number of customers place a large number of orders. This suggests that customer engagement varies significantly across the population, with some customers being highly active and others having lower levels of engagement. The presence of this skewed distribution indicates that customer order frequency does not follow a normal distribution and instead represents count data with variability across customers. This pattern is typical in transactional datasets and supports the use of count regression models such as Poisson and Negative Binomial regression.

proportion	
churn	
0	0.693583
1	0.306417
dtype: float64	

Figure 1.5.2: Distribution of Customer Churn

Figure 1.5.2 shows the distribution of customers classified as churned and retained based on the recency threshold used in this study. Approximately 30.6% of customers were classified as churned, while 69.4% remained active, indicating that most customers continued to engage with the platform. This distribution confirms the presence of both churned and active customers, which is necessary for training classification models. Understanding this distribution is important because it provides insight into customer retention patterns and ensures that predictive models such as logistic regression, decision trees, and random forests can effectively identify customers at risk of churn based on their behavioral features.

Over-Dispersion Analysis

Customer order frequency was examined to determine whether it satisfies the assumptions of Poisson regression. The mean order frequency was calculated to be approximately 16.59, while the variance was approximately 277.38. Since the variance is significantly larger than the mean, the data exhibits over-dispersion. Over-dispersion violates the assumptions of Poisson regression, which assumes equal mean and variance. Therefore, Negative Binomial regression, which accounts for over-dispersion, was selected as a more appropriate model for usage intensity. This analysis ensures that the selected statistical model accurately represents the underlying data distribution.

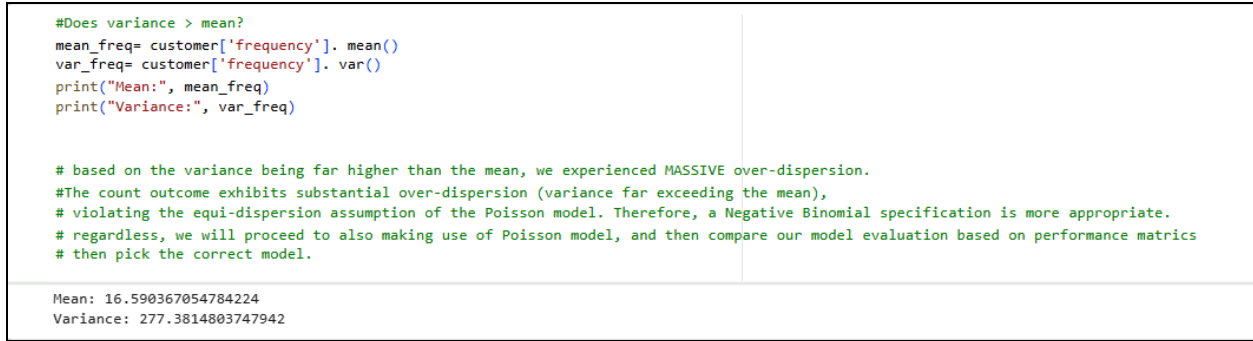


Figure 1.5.3: Mean and Variance of Customer Order Frequency

Figure 1.5.3 shows the calculated mean and variance of customer order frequency. The mean frequency was approximately 16.59, while the variance was significantly higher at approximately 277.38. This indicates that the variance is much greater than the mean, demonstrating substantial over-dispersion in the data. Over-dispersion occurs when variability in the data exceeds the assumptions of the Poisson distribution, which requires the mean and variance to be equal. This result is important because it determines the appropriate statistical model for usage intensity analysis. Since the equi-dispersion assumption of Poisson regression is violated, the Negative Binomial regression model, which accounts for over-dispersion, is more suitable for modeling customer order frequency. This analysis ensures that the selected statistical model accurately represents the underlying distribution of customer behavior.

2. Analysis Part I: Customer Churn Modeling

In this section, we analyze customer churn using classification models based on behavioral features derived from the Instacart transactional dataset. Customer churn is defined using the recency variable, where customers with higher recency values are considered inactive and classified as churned. Predicting customer churn is important because it enables businesses to identify customers at risk of disengagement and implement strategies to improve retention.

Specifically, the following models and analyses will be discussed:

- Data preparation and train-test split
- Logistic Regression Classification
- Decision Tree Classification
- Random Forest Classification
- Model comparison using Accuracy and AUC
- Model evaluation using Confusion Matrix, Classification Report, and ROC Curve

These models were selected to compare both statistical and machine learning approaches to classification. Logistic regression serves as a baseline statistical model, while decision tree and

random forest models capture more complex relationships between customer behavioral features and churn outcomes.

2.1 Data Preparation for Classification Models

Before training our classification models, the dataset was prepared by selecting predictor variables and splitting the data into training and testing sets. The predictor variables used were frequency, total_active_days, and avg_days_between_orders. The dataset was split into training and testing sets using a 70%–30% split. Stratified sampling was used to preserve the proportion of churned and retained customers in both datasets.

In this section, we apply several classification models to predict customer churn based on behavioral features. Logistic regression was selected as the baseline model because it is widely used for binary classification problems and provides interpretable relationships between predictor variables and the probability of churn

Additionally, decision tree and random forest models were implemented to capture potential nonlinear relationships and interactions between customer behavioral variables that may not be fully captured by logistic regression. These models allow for a comprehensive comparison of linear and nonlinear approaches to churn prediction.

```
# Data preparation for modeling
from sklearn.model_selection import train_test_split
X= customer[['frequency',
             'total_active_days',
             'avg_days_between_orders']]
y= customer['churn']
X_train, X_test, y_train, y_test= train_test_split(
    X, y,
    test_size=0.3,
    random_state=42,
    stratify=y # keeps churn proportion balanced
)
# we stratified Because churn is 30%. We want that preserved in both sets.
```

Figure 2.1.1: Train-Test Split for Customer Churn Modeling

2.2 Logistic Regression

Logistic regression was used as the baseline statistical model for predicting customer churn. Logistic regression estimates the probability of a binary outcome using a logistic function and is widely used for classification problems. This model is particularly useful because it provides

interpretable results and establishes a baseline for comparing more complex machine learning models.

```
Logistic AUC: 0.810808016162468
Logistic Accuracy: 0.7644634110857863
```

Figure 2.2.1: Logistic Regression Model Performance

Figure 2.2.1 shows the performance of the logistic regression model. The model achieved an accuracy of 0.764 and an AUC of 0.811, indicating strong predictive performance. The AUC value measures the model's ability to distinguish between churned and retained customers, and a value above 0.80 indicates good classification ability.

Confusion Matrix:				
[[37593 5314]				
[9257 9699]]				
Classification Report:				
	precision	recall	f1-score	support
0	0.80	0.88	0.84	42907
1	0.65	0.51	0.57	18956
accuracy			0.76	61863
macro avg	0.72	0.69	0.70	61863
weighted avg	0.75	0.76	0.76	61863

Figure 2.2.2: Logistic Regression Confusion Matrix and Classification Report

Figure 2.2.2 shows the confusion matrix and classification report for the logistic regression model. The confusion matrix indicates that the model correctly classified a large number of retained customers and successfully identified a substantial portion of churned customers. The classification report shows precision, recall, and F1-score values, indicating balanced performance across both classes.

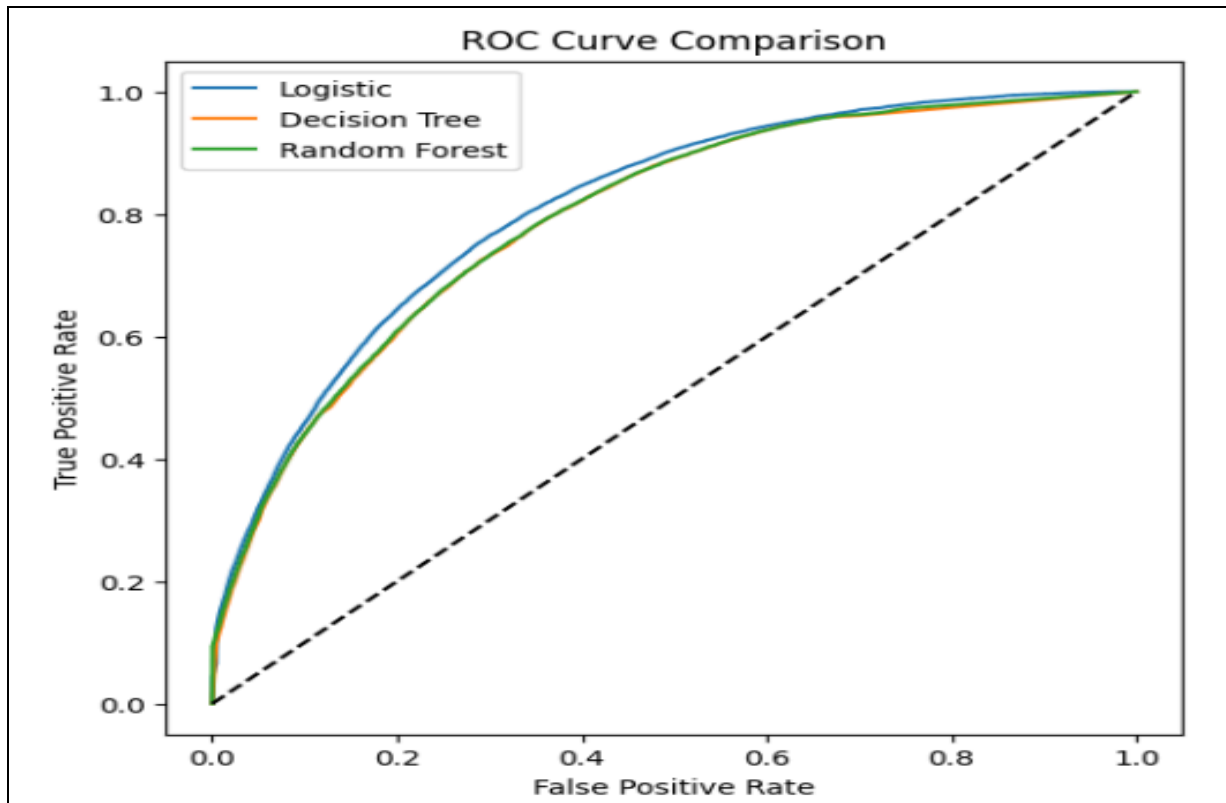


Figure 2.2.3: ROC Curve for Logistic Regression Model

Figure 2.2.3 shows the ROC curve, which plots the true positive rate against the false positive rate at different classification thresholds. The curve demonstrates strong separation between churned and retained customers, confirming that logistic regression provides effective predictive performance.

These results indicate that logistic regression effectively captures the relationship between customer behavioral features and churn probability. The model demonstrates strong predictive performance, as reflected by both accuracy and AUC values. This suggests that behavioral features such as order frequency, recency, and activity duration provide meaningful information for identifying customers at risk of churn.

2.3 Decision Tree Classification

Decision tree classification was used as a non-linear classification model to predict customer churn. Decision trees classify observations by recursively splitting the dataset based on predictor variables. This method is useful because it captures complex relationships between behavioral features and churn outcomes.

```
from sklearn.tree import DecisionTreeClassifier
tree_model = DecisionTreeClassifier(random_state=42)
tree_model.fit(X_train, y_train)
tree_pred = tree_model.predict(X_test)
tree_prob = tree_model.predict_proba(X_test)[:, 1]
tree_auc = roc_auc_score(y_test, tree_prob)
tree_acc = accuracy_score(y_test, tree_pred)
print("Decision Tree AUC:", tree_auc)
print("Decision Tree Accuracy:", tree_acc)
```

```
Decision Tree AUC: 0.7929362509757419
Decision Tree Accuracy: 0.7583854646557716
```

Figure 2.3.1: Decision Tree Model Performance

From Figure 2.3.1, we observe that the decision tree model achieved an accuracy of 0.758 and an AUC of 0.793, indicating strong classification performance. Although these values are slightly lower compared to the logistic regression model, the decision tree still performs effectively in predicting customer churn and capturing important behavioral patterns.

The decision tree model provides a flexible approach for capturing nonlinear relationships between customer behavioral features and churn. Although its predictive performance was slightly lower than logistic regression, it offers valuable insights into decision boundaries and behavioral patterns that influence churn risk.

2.4 Random Forest Classification

Random forest classification was used as an ensemble learning method to predict customer churn. Random forest builds multiple decision trees and combines their predictions to improve classification accuracy and reduce overfitting. This approach is particularly useful for capturing complex relationships between predictor variables and improving prediction stability.

Although the random forest model performs slightly better than the decision tree model in terms of AUC, its performance remains slightly lower than logistic regression. Nevertheless, random forest provides reliable predictions and is capable of capturing more complex relationships between customer engagement variables and churn outcomes.

```
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
rf_model.fit(X_train, y_train)
rf_pred = rf_model.predict(X_test)
rf_prob = rf_model.predict_proba(X_test)[:, 1]
rf_auc = roc_auc_score(y_test, rf_prob)
rf_acc = accuracy_score(y_test, rf_pred)
print("Random Forest AUC:", rf_auc)
print("Random Forest Accuracy:", rf_acc)
```

```
Random Forest AUC: 0.7952028134940783
Random Forest Accuracy: 0.7576095566008761
```

Figure 2.4.1: Random Forest Model Performance

From Figure 2.4.1, we observe that the random forest model achieved an accuracy of 0.758 and an AUC of 0.795, indicating strong predictive performance. These results demonstrate that the model is effective in distinguishing between churned and retained customers based on behavioral features.

The random forest model improves upon the decision tree by combining multiple trees to reduce variance and improve generalization. This ensemble approach captures complex relationships between behavioral features and churn while reducing the risk of overfitting. Although its performance was slightly lower than logistic regression in this case, it remains a powerful and robust predictive model.

2.5 Model Comparison

In this subsection, we compare the performance of the logistic regression, decision tree, and random forest models to determine which model provides the best predictive performance for customer churn. The models were evaluated using accuracy and Area Under the ROC Curve (AUC), which measure the overall classification accuracy and the model's ability to distinguish between churned and retained customers.

Comparing multiple models is important because different classification algorithms capture patterns in data differently.

	Model	AUC	Accuracy
0	Logistic	0.810808	0.764463
1	Decision Tree	0.792936	0.758385
2	Random Forest	0.795203	0.757610

Figure 2.5.1: Comparison of Classification Model Performance

From Figure 2.5.1, we observe that the logistic regression model achieved the highest accuracy of 0.764 and the highest AUC of 0.811 among the three models. This indicates that logistic regression provides the best overall performance in predicting customer churn. The decision tree and random forest models also demonstrated strong predictive performance, but their accuracy and AUC values were slightly lower than logistic regression.

3 Analysis Part II: Usage Intensity Modeling

In this section, we analyze customer usage intensity using count regression models based on customer behavioral features derived from the Instacart transactional dataset. Usage intensity is defined as the total number of orders placed by each customer and represents an important measure of customer engagement. Modeling usage intensity allows us to understand how customer behavioral features influence ordering frequency and helps predict future engagement levels.

Unlike churn modeling, which involves binary classification, usage intensity modeling involves predicting count data. Count data consists of non-negative integer values and requires specialized regression models such as Poisson and Negative Binomial regression. As discussed in Section 1.5, the variance of customer order frequency was significantly greater than the mean, indicating the presence of over-dispersion. This violates the assumptions of Poisson regression and suggests that Negative Binomial regression may provide a better fit.

Specifically, the following models and analyses will be discussed:

- Poisson Regression
- Negative Binomial Regression
- Model comparison using RMSE
- Final model selection

These models were selected to evaluate how well each statistical approach models customer usage intensity and to determine the most appropriate model for predicting customer ordering behavior.

3.1 Poisson Regression

In this subsection, we fit a Poisson regression model to analyze customer usage intensity. Poisson regression is a commonly used statistical method for modeling count data, where the response variable represents the number of occurrences of an event. In this study, the response variable is frequency, which represents the total number of orders placed by each customer.

Poisson regression is commonly used for modeling count data where the response variable represents the number of occurrences of an event (James, Witten, Hastie, & Tibshirani, 2021). The predictor variables used in this model were `total_active_days`, `avg_days_between_orders`, and `recency`. These variables capture important aspects of customer engagement, including how long customers remain active, how frequently they place orders, and how recently they interacted with the platform.

Poisson regression assumes that the mean and variance of the count variable are equal, a property known as equi-dispersion. This assumption is important because violation of equi-dispersion can result in biased parameter estimates and unreliable inference. Therefore, it is necessary to examine whether the variance of customer order frequency is consistent with the Poisson assumption before determining the suitability of the model.

We implemented the Poisson regression model using the Generalized Linear Model (GLM) framework with a log link function. The predictor variables included `total_active_days`, `avg_days_between_orders`, and `recency`, which capture important aspects of customer engagement and ordering behavior. These variables were selected because they provide meaningful information about customer activity duration, ordering consistency, and recent engagement. By fitting the Poisson regression model, we aim to understand the relationship between these behavioral features and customer usage intensity.

Generalized Linear Model Regression Results						
=====						
Dep. Variable:	frequency	No. Observations:	206209			
Model:	GLM	Df Residuals:	206205			
Model Family:	Poisson	Df Model:	3			
Link Function:	Log	Scale:	1.0000			
Method:	IRLS	Log-Likelihood:	-5.3680e+05			
Date:	Thu, 12 Feb 2026	Deviance:	1.8467e+05			
Time:	15:04:06	Pearson chi2:	2.05e+05			
No. Iterations:	5	Pseudo R-squ. (CS):	1.000			
Covariance Type:	nonrobust					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	2.6468	0.002	1278.277	0.000	2.643	2.651
total_active_days	0.0056	5.26e-06	1068.219	0.000	0.006	0.006
avg_days_between_orders	-0.0843	0.000	-600.117	0.000	-0.085	-0.084
recency	-0.0005	6.85e-05	-7.037	0.000	-0.001	-0.000
=====						
Poisson AIC: 1073616.8756247978						

Figure 3.1.1: Poisson Regression Model Results

From the output (Figure 3.1.1), we can observe the estimated regression coefficients and statistical significance of the predictor variables in the Poisson regression model. The variable `total_active_days` has a positive coefficient, indicating that customers who remain active on the platform for longer periods tend to place more orders. In contrast, the variables `avg_days_between_orders` and `recency` have negative coefficients, suggesting that customers who have longer gaps between orders or who have not placed recent orders tend to have lower order frequency. All predictor variables are statistically significant, indicating that they have meaningful relationships with customer usage intensity.

3.2 Negative Binomial Regression

In this subsection, we apply the Negative Binomial regression model to analyze customer usage intensity. Negative Binomial regression is an extension of Poisson regression that is specifically designed to handle over-dispersed count data, where the variance exceeds the mean. As discussed earlier in the exploratory data analysis, the variance of customer order frequency was significantly greater than the mean, indicating the presence of over-dispersion. Negative Binomial regression extends Poisson regression by accounting for over-dispersion in count data (James, Witten, Hastie, & Tibshirani, 2021). Therefore, the Negative Binomial regression model is expected to provide a better fit for the data compared to the Poisson regression model.

Similar to the Poisson regression model, the predictor variables used in this analysis were `total_active_days`, `avg_days_between_orders`, and `recency`. The Negative Binomial model was implemented using Maximum Likelihood Estimation (MLE), which allows for accurate estimation of model parameters while accounting for over-dispersion. This approach enables us to

examine the relationship between customer behavioral features and order frequency more reliably. The results of the Negative Binomial regression model are presented below.

In addition to coefficient interpretation, the Negative Binomial model estimates a dispersion parameter (alpha), which quantifies the degree of over-dispersion in the data. The statistically significant dispersion parameter confirms that the variance exceeds the mean, validating the presence of over-dispersion. Furthermore, the Negative Binomial model achieved a substantially lower Akaike Information Criterion (AIC) compared to the Poisson model, indicating a better overall statistical fit. These results confirm that the Negative Binomial regression model provides a more appropriate and reliable framework for modeling customer usage intensity.

NegativeBinomial Regression Results						
=====						
Dep. Variable:	frequency	No. Observations:	206209			
Model:	NegativeBinomial	Df Residuals:	206205			
Method:	MLE	Df Model:	3			
Date:	Thu, 12 Feb 2026	Pseudo R-squ.:	0.3196			
Time:	15:04:24	Log-Likelihood:	-5.3041e+05			
converged:	False	LL-Null:	-7.7953e+05			
Covariance Type:	nonrobust	LLR p-value:	0.000			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	2.5645	0.002	1068.905	0.000	2.560	2.569
total_active_days	0.0058	6.23e-06	924.472	0.000	0.006	0.006
avg_days_between_orders	-0.0799	0.000	-497.041	0.000	-0.080	-0.080
recency	-0.0005	7.81e-05	-6.265	0.000	-0.001	-0.000
alpha	0.0136	0.000	82.915	0.000	0.013	0.014

Figure 3.2.1: Negative Binomial Regression Model Results

From the output (Figure 3.2.1), we can see that total_active_days has a positive coefficient, indicating that customers who remain active longer tend to place more orders. The variables avg_days_between_orders and recency have negative coefficients, suggesting that customers with longer gaps between orders or less recent activity tend to have lower order frequency. All predictor variables are statistically significant, indicating that they have an important influence on customer usage intensity.

Additionally, the model estimates a dispersion parameter (alpha), which confirms the presence of over-dispersion in the data. This indicates that the Negative Binomial regression model is more appropriate than the Poisson regression model for modeling customer usage intensity, as it accounts for variability that is not captured by the Poisson model.

3.3 Model Comparison

In this subsection, we compare the performance of the Poisson regression and Negative Binomial regression models to determine which model provides the best fit for modeling customer usage intensity. Both models were evaluated using Akaike Information Criterion (AIC) and Root Mean Squared Error (RMSE). The AIC measures model fit while penalizing model complexity, with lower values indicating a better model. RMSE measures the difference between observed and predicted values, with lower values indicating more accurate predictions.

```
Poisson RMSE: 5.444765647309177  
Negative Binomial RMSE: 6.311388328563368
```

Figure 3.3.1: RMSE Comparison of Poisson and Negative Binomial Models

From the output (Figure 3.3.1), we can see that the Poisson regression model achieved an RMSE of 5.445, while the Negative Binomial regression model achieved an RMSE of 6.311. This indicates that the Poisson regression model produced predictions that were closer to the observed values compared to the Negative Binomial model.

However, despite the lower RMSE value, the Poisson regression model assumes that the mean and variance of the response variable are equal. As shown earlier in the exploratory data analysis, the variance of customer order frequency was significantly greater than the mean, indicating over-dispersion. This violates the assumptions of the Poisson model and suggests that the Negative Binomial model provides a more appropriate statistical representation of the data.

Although the Poisson model achieved slightly lower RMSE, RMSE alone is not sufficient to determine the best model for count data. The Poisson model assumes equi-dispersion, which was violated in this dataset due to significant over-dispersion. In contrast, the Negative Binomial model accounts for this variability through the dispersion parameter and achieved a substantially lower AIC value, indicating a superior statistical fit. Therefore, despite the slightly higher RMSE, the Negative Binomial regression model is more appropriate for modeling customer usage intensity.

4. Discussion

The results of this study demonstrate that customer behavioural features derived from transactional data provide meaningful insight into both customer retention and usage intensity. Across the churn modelling task, logistic regression achieved the highest classification performance while maintaining interpretability, illustrating that relatively simple statistical models can perform competitively when behavioural predictors are carefully engineered.

One key finding is that behavioural consistency plays a central role in customer engagement. Customers with higher order frequency and shorter gaps between purchases exhibited lower churn risk and greater usage intensity. This aligns with behavioural theory in customer analytics, where sustained interaction patterns indicate stronger platform attachment.

The comparison between statistical and machine learning approaches provides additional insight. Although decision tree and random forest models captured nonlinear relationships, logistic regression demonstrated superior AUC performance. This suggests that the relationship between behavioural features and churn risk is largely monotonic and can be effectively captured through interpretable statistical models.

For usage intensity modelling, the exploratory analysis revealed substantial over-dispersion in customer order frequency, with variance far exceeding the mean. While Poisson regression achieved slightly lower RMSE, the violation of equi-dispersion assumptions indicated that it was not statistically appropriate. The Negative Binomial regression model accounted for this additional variability and provided more reliable parameter estimates. This highlights the importance of prioritizing distributional validity over predictive accuracy alone when selecting statistical models.

One limitation of this analysis is that only behavioral timing features were used to model customer churn and usage intensity. Additional features such as product categories, order size, customer demographics, and purchase value could further improve model performance. Furthermore, more advanced machine learning methods, such as gradient boosting or neural networks, may provide higher predictive accuracy. Future research could incorporate these additional features and modeling approaches to enhance predictive performance and provide deeper insights into customer behavior.

Overall, the findings emphasize that model selection should be guided by both statistical assumptions and interpretability considerations. Behavioral engagement features provide a robust foundation for modelling customer outcomes, and selecting models aligned with data structure leads to more reliable insights.

5. Summary of Findings

The analysis produced several key findings related to customer retention and usage intensity.

First, logistic regression achieved approximately 76% classification accuracy and an AUC of 0.811, demonstrating strong predictive performance in identifying customer churn risk. The model performed particularly well in identifying retained customers, while churn detection remained more challenging, reflecting the inherent complexity of predicting customer inactivity.

Second, exploratory analysis confirmed that customer order frequency exhibits substantial over-dispersion, with variance significantly exceeding the mean. This finding motivated the

transition from Poisson regression to Negative Binomial regression for modelling usage intensity.

Third, although the Poisson model achieved slightly lower RMSE, the Negative Binomial regression model was selected as the final model due to its lower AIC and its ability to account for over-dispersion. This decision reflects a statistically grounded approach to model selection.

Finally, the results demonstrate that behavioural features such as frequency, engagement duration, and ordering consistency provide strong signals for predicting both churn and usage intensity. The selected models balance predictive performance with interpretability, making them suitable for practical applications.

6. Limitations

While the modelling framework demonstrates strong statistical performance, several methodological considerations should be acknowledged

Customer churn in this study was inferred using recency thresholds rather than directly observed behavioural outcomes. Although this approach provides a reasonable proxy for inactivity, it may not fully capture true disengagement patterns.

Additionally, the analysis relied on aggregated behavioural features derived from transactional data. Product-level information, spending behaviour, and external contextual variables were not included, which may limit the ability to capture more nuanced customer preferences.

Temporal dynamics were also simplified. The dataset does not contain explicit calendar timestamps, which prevents modelling of seasonality or longitudinal behavioural trends.

Furthermore, the models identify statistical associations rather than causal relationships. External factors influencing customer behaviour, such as pricing or marketing interventions, were not considered.

Finally, the findings are specific to the Instacart dataset and may not generalize directly to other platforms or industries.

7. Future Work

Future research could extend this analysis in several ways.

Incorporating product-level and monetary variables could provide deeper insight into customer preferences and purchasing patterns. Integrating basket composition or spending behaviour may improve predictive performance.

Temporal modelling approaches, such as survival analysis or time-to-event modelling, could provide a more refined understanding of churn dynamics by explicitly modelling the timing of customer inactivity.

Advanced modelling techniques, including interaction effects, nonlinear models, or hierarchical modelling frameworks, could further capture heterogeneity across customer segments.

Additionally, cost-sensitive classification approaches could be explored to optimize decision thresholds based on business objectives, particularly in scenarios where false negatives or false positives carry different operational costs.

8. Conclusion

This study applied statistical and machine learning methods to analyze customer retention and usage intensity using transactional data from the Instacart online grocery platform. Behavioural features derived from ordering patterns provided meaningful signals for predicting both churn risk and order frequency.

Among the classification models evaluated, logistic regression demonstrated the strongest balance between predictive performance and interpretability, achieving an accuracy of approximately 0.764 and an AUC of 0.811. The results indicate that customer engagement patterns can be effectively captured through interpretable statistical models.

For usage intensity modelling, exploratory analysis revealed substantial over-dispersion in customer order frequency. Although Poisson regression provided competitive predictive accuracy, its underlying assumptions were violated. The Negative Binomial regression model accounted for this over-dispersion and was selected as the preferred statistical model due to its lower AIC and more appropriate distributional assumptions.

Overall, the findings demonstrate that customer behavioural patterns derived from transactional data can be effectively captured through statistically grounded modelling approaches. Logistic regression provided an interpretable and robust framework for churn prediction, while Negative Binomial regression addressed the structural over-dispersion present in customer order frequency. By aligning model choice with the statistical characteristics of the data and the objectives of behavioural analysis, this study contributes meaningful insight into customer

engagement dynamics and highlights the broader importance of selecting models that balance interpretability, statistical validity, and predictive performance.

References

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Appendix

This appendix contains the Python code used for data preprocessing, feature engineering, customer churn modeling, usage intensity modeling, and model comparison. These codes were used to prepare the dataset, construct predictive models, and evaluate their performance.

Colab Notebook

Python Code

Appendix A: Pre-Processing of the Data

A.1 Load Google Drive (Colab)

```
from google.colab import drive
drive.mount('/content/drive')
```

A.2 Reading the Dataset and Inspecting the structure

```
import pandas as pd
```

```
orders = pd.read_csv("/content/drive/MyDrive/DATA 606/orders.csv")
```

```
print(orders.shape)
print(orders.columns)
print(orders.head())
print(orders.isnull().sum())
```

A.3 Handling Missing Values

```
orders['days_since_prior_order'] = orders['days_since_prior_order'].fillna(0)
```

A.4 Confirming Data Types

```
orders.dtypes
```

A.5 Dropping Unused Columns and Summarizing Numeric Variables

```
orders = orders.drop(columns=['order_id', 'eval_set'])
orders.describe()
```

Appendix B: Feature Engineering

B.1 Aggregating Order Data to Customer-level Features

```
# Aggregation
customer = orders.groupby('user_id').agg(
    frequency=('order_number', 'max'),
    total_active_days=('days_since_prior_order', 'sum'),
    avg_days_between_orders=('days_since_prior_order', 'mean')
).reset_index()

customer.shape
customer.head()
```

B.2 Creating Recency Feature (last observed days_since_prior_order)

```
#create recency
last_orders = orders.sort_values(['user_id', 'order_number']) \
    .groupby('user_id') \
    .tail(1)[['user_id', 'days_since_prior_order']]

last_orders = last_orders.rename(columns={'days_since_prior_order': 'recency'})

customer = customer.merge(last_orders, on='user_id', how='left')
```

B.3 Defining Churn Based on a Threshold (75th percentile of recency)

```
threshold = customer['recency'].quantile(0.75)
customer['churn'] = (customer['recency'] >= threshold).astype(int)
```

Appendix C: Exploratory Data Analysis

C.1 Summary Statistics of Customer Features

```
customer.describe()
```

C.2 Distribution of Customer Frequency

```
customer['frequency'].hist(bins=50)
```

C.3 Distribution of Churn Classes

```
customer['churn'].value_counts(normalize=True)
```

C.4 Over-Dispersion Check (Variance vs Mean of Frequency)

```
#Does variance > mean?
mean_freq= customer['frequency']. mean()
var_freq= customer['frequency']. var()
print("Mean:", mean_freq)
print("Variance:", var_freq)
```

#Based on the variance being far higher than the mean, we experienced MASSIVE over-dispersion.

#The count outcome exhibits substantial over-dispersion (variance far exceeding the mean), #violating the equi-dispersion assumption of the Poisson model. Therefore, a Negative Binomial specification is more appropriate.

#Regardless, we will proceed to also make use of the Poisson regression model, and then compare our model evaluation based on performance metrics, and then select the correct model.

Appendix D: Customer Churn Modeling (Classification)

D.1 Train-test split (stratified)

```
# Data preparation for modeling
from sklearn.model_selection import train_test_split
X= customer[['frequency',
             'total_active_days',
             'avg_days_between_orders']]
y= customer['churn']
X_train, X_test, y_train, y_test= train_test_split(
    X, y,
    test_size=0.3,
```

```
random_state=42,  
stratify=y )
```

D.2 Logistic Regression (Baseline Statistical Model)

```
from sklearn.linear_model import LogisticRegression  
from sklearn.metrics import roc_auc_score, accuracy_score  
log_model = LogisticRegression(max_iter=1000)  
log_model.fit(X_train, y_train)
```

```
log_pred = log_model.predict(X_test)  
log_prob = log_model.predict_proba(X_test)[:, 1]
```

```
log_auc = roc_auc_score(y_test, log_prob)  
log_acc = accuracy_score(y_test, log_pred)
```

```
print("Logistic AUC:", log_auc)  
print("Logistic Accuracy:", log_acc)
```

D.3 Decision Tree

```
from sklearn.tree import DecisionTreeClassifier  
tree_model = DecisionTreeClassifier(random_state=42)  
tree_model.fit(X_train, y_train)  
tree_pred = tree_model.predict(X_test)  
tree_prob = tree_model.predict_proba(X_test)[:, 1]  
tree_auc = roc_auc_score(y_test, tree_prob)  
tree_acc = accuracy_score(y_test, tree_pred)  
print("Decision Tree AUC:", tree_auc)  
print("Decision Tree Accuracy:", tree_acc)
```

D.4 Random Forest

```
from sklearn.ensemble import RandomForestClassifier  
rf_model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)  
rf_model.fit(X_train, y_train)  
rf_pred = rf_model.predict(X_test)  
rf_prob = rf_model.predict_proba(X_test)[:, 1]  
rf_auc = roc_auc_score(y_test, rf_prob)  
rf_acc = accuracy_score(y_test, rf_pred)  
print("Random Forest AUC:", rf_auc)  
print("Random Forest Accuracy:", rf_acc)
```

D.5 Model Performance Table

```
import pandas as pd

results = pd.DataFrame({
    "Model": ["Logistic", "Decision Tree", "Random Forest"],
    "AUC": [log_auc, tree_auc, rf_auc],
    "Accuracy": [log_acc, tree_acc, rf_acc]
})

results
```

D.6 ROC Curve Comparison Plot

```
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve

plt.figure(figsize=(6,6))

for name, prob in [
    ("Logistic", log_prob),
    ("Decision Tree", tree_prob),
    ("Random Forest", rf_prob)
]:
    fpr, tpr, _ = roc_curve(y_test, prob)
    plt.plot(fpr, tpr, label=name)

plt.plot([0,1],[0,1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")
plt.legend()
plt.show()
```

D.7 Confusion Matrix and Classification Report (Logistic Regression)

```
from sklearn.metrics import confusion_matrix, classification_report

cm = confusion_matrix(y_test, log_pred)
print("Confusion Matrix:\n", cm)

print("\nClassification Report:\n")
print(classification_report(y_test, log_pred))
```

Appendix E: Usage Intensity Modeling (Count Models)

E.1 Preparing predictors and response for count regression

```
import statsmodels.api as sm

X_count = customer[['total_active_days',
                    'avg_days_between_orders',
                    'recency']]

X_count = sm.add_constant(X_count)

y_count = customer['frequency']
```

E.2 Poisson Regression

```
poisson_model = sm.GLM(y_count, X_count,
                       family=sm.families.Poisson()).fit()

print(poisson_model.summary())
print("Poisson AIC:", poisson_model.aic)
```

E.3 Negative Binomial Regression

```
nb_model = sm.GLM(y_count, X_count,
                  family=sm.families.NegativeBinomial()).fit()

print(nb_model.summary())
print("Negative Binomial AIC:", nb_model.aic)
```

E.4 RMSE comparison

```
#compare predictive model

from sklearn.metrics import mean_squared_error
import numpy as np
# RMSE
# Poisson predictions
poisson_pred = poisson_model.predict(X_count)

# NB predictions
nb_pred = nb_model.predict(X_count)

poisson_rmse = np.sqrt(mean_squared_error(y_count, poisson_pred))
nb_rmse = np.sqrt(mean_squared_error(y_count, nb_pred))
```

```
print("Poisson RMSE:", poisson_rmse)
print("Negative Binomial RMSE:", nb_rmse)
```

E.5 Negative Binomial Regression for Discrete MLE and Summary

```
from statsmodels.discrete.discrete_model import NegativeBinomial
import statsmodels.api as sm
```

```
X_nb = customer[['total_active_days',
                 'avg_days_between_orders',
                 'recency']]
```

```
X_nb = sm.add_constant(X_nb)
y_nb = customer['frequency']
```

```
nb2_model = NegativeBinomial(y_nb, X_nb).fit()
```

```
print(nb2_model.summary())
```

E.6 Proper NB AIC + repeated summary (as in notebook)

```
print("Proper NB AIC:", nb2_model.aic)
```

```
print(nb2_model.summary())
```